Multi-Asset Portfolio Risk Framework

Complete Output Interpretation Guide

Purpose: For every numerical output and every plot in the project, this guide explains exactly what you are looking at, how to read it, what the numbers mean in plain English, what good and bad results look like, and why each output matters.

How to use this: Before an interview, read through the section for each phase. You should be able to explain any output on your screen in under 60 seconds.

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Phase 1 — Data & Portfolio Setup

OUTPUT 1.1 — Shape After `dropna`

What you are looking at:
Two numbers printed after downloading price data: shape before and after dropping rows with missing values.

```
Shape before dropna : (2517, 8)
Shape after  dropna : (2515, 8)
```

How to read it:

- First number = total trading days downloaded

- Second number = trading days with complete data across all 8 assets
- The difference = days dropped due to at least one asset having a missing price

What it means in plain English:
You are checking that the dataset is clean. If SPY has data on a day but GLD does not, that entire row gets removed. You only keep days where you have complete information for all 8 assets simultaneously.

Good result:

- ~2,500 rows for a 2015–2024 date range (roughly 252 trading days × 10 years)
- Fewer than 10 rows dropped — small gaps from market holidays or ETF-specific non-trading days are normal

Warning sign:

- More than 50 rows dropped — suggests a data quality issue with one of the tickers. Run `raw.isna().sum()` per column to identify which asset has missing data.
- Fewer than 2,000 rows — suggests the date range was not downloaded correctly

Why it matters downstream:
The covariance matrix, all VaR calculations, and all backtesting are computed on this dataset. A dirty dataset with gaps would corrupt the covariance matrix and produce silently wrong VaR numbers.

OUTPUT 1.2 — Log Returns: First 5 Rows

What you are looking at:
A table showing the daily log return for each of the 8 assets on the first 5 trading days after the start date.

How to read it:

- Each row = one trading day
- Each column = one asset's log return on that day
- Values are in decimal form: 0.0123 means a +1.23% return; -0.0056 means a -0.56% return

What it means in plain English:
On any given day, each number tells you the percentage gain or loss on that asset. A log return of 0.01 means the asset price rose by approximately 1% from the previous day. Negative numbers mean the price fell.

Good result:

- Values in the range of roughly -0.03 to +0.03 on typical days (i.e., within ±3%)
- No NaN values in any cell

- The first date is January 2, 2015 (or the first trading day after your start date) — the first row of prices gets consumed computing the first return

Warning sign:

- Any value outside ±0.20 (±20%) on a single day — this almost certainly indicates a data error (unadjusted split, bad yfinance data point)
- NaN values — means `dropna()` failed or was not applied

Why it matters downstream:
All VaR, CVaR, covariance, and portfolio return calculations are built directly on this returns DataFrame. A single corrupted return on one day will distort every risk metric.

OUTPUT 1.3 — Sanity Check: `|return| > 20%`

What you are looking at:
A count, per asset, of how many days had an absolute daily return exceeding 20%.

```
Days with |return| > 20%:
SPY    0
QQQ    0
EFA    0
GLD    0
TLT    0
LQD    0
XOM    0
VNQ    0
```

How to read it:
Zero for every asset means no suspicious data points were found.

Good result:
All zeros. A 20% single-day move in any of these diversified ETFs would be extraordinary and almost certainly indicates a data error.

Warning sign:
Any non-zero value. If you see, for example, `TLT: 1`, investigate that specific date. Run `log_returns[log_returns['TLT'].abs() > 0.20]` to find it. It is almost always an unadjusted dividend or split that yfinance did not handle correctly for that specific pull.

Why it matters:
A single artificially extreme return in TLT, for example, would dramatically inflate the TLT variance and

distort the covariance matrix. The 99% VaR would be wrong because it would be partly calibrated to a data error.

PLOT 1.4 — Normalised Price Chart (Base = 100)

What you are looking at:
A line chart where all 8 assets start at 100 on January 1, 2015, and grow or shrink from there. By the end of the chart, the y-value for each asset shows its total return multiple from the starting date.

How to read it:

- Every asset starts at 100 (even though their actual prices are different — SPY was ~$200, GLD was ~$115)
- A value of 200 at the end means the asset doubled from 2015 to 2024
- A value of 80 means the asset lost 20% over the entire period
- The chart shows relative performance, not absolute price

What specific things to look for:

SPY and QQQ should be among the top performers — strong equity bull market from 2015 to 2021, then a correction in 2022.

TLT should be interesting — it rallied strongly through 2020 (flight to safety during COVID) then collapsed through 2022 as the Fed raised rates aggressively. By 2024 TLT may still be below its 2015 starting value.

GLD should show mild positive performance with a spike during 2020.

VNQ should show solid gains through 2021 then a sharp drop in 2022 as rising rates hit real estate.

Good result:
Clear divergence across asset classes. If all lines move together identically, your portfolio has no real diversification — every asset responds to the same factor.

Warning sign:
All assets moving in perfect lockstep throughout — suggests your data may have a column alignment error (same data in multiple columns). Or, it could simply mean you chose highly correlated assets, which is a portfolio construction problem, not a data problem.

Why it matters:
This chart is your first visual confirmation that the portfolio is genuinely diversified. If TLT and GLD hold their value while SPY and QQQ fall during the 2020 and 2022 drawdowns, diversification is working. This will be reflected in the covariance matrix and ultimately determines how much the portfolio VaR is reduced relative to any single-asset VaR.

PLOT 1.5 — Portfolio Daily Return (Fill Chart)

What you are looking at:
A time series of the portfolio's daily log returns, with gains shown in blue fill above zero and losses in red fill below zero.

How to read it:

- The y-axis is in percentage terms (already multiplied by 100)
- Each vertical spike is one trading day
- Blue spikes above zero = profitable days; red spikes below zero = loss days
- The height of each spike = the magnitude of that day's return

What specific things to look for:

- **March 2020:** You should see a sharp cluster of large red spikes — the COVID crash. The worst days should show losses of 4–8% in a single day.
- **2022:** A prolonged period with more red than blue — the rate shock year.
- **2017–2019:** Mostly shallow spikes with very few large red days — the low-volatility pre-COVID period.

Good result:
The distribution of spike heights is uneven — most days have small moves (under 1%), with rare but very large spikes on both sides. This is the characteristic "thin middle, fat tail" shape of financial returns.

Warning sign:
If the spikes are uniformly sized throughout (constant volatility), that is suspicious — real financial data has volatility clustering. It suggests you may have made an error in the return computation (e.g., accidentally used daily price levels instead of returns).

Why it matters:
The return series is what every downstream model is built on. The visual clustering of large spikes in certain periods (COVID, 2022) is precisely what the Christoffersen backtesting test will detect — exceptions are not randomly distributed in time.

PLOT 1.6 — Return Distribution vs Normal Overlay

What you are looking at:
A histogram of all daily portfolio returns (blue bars), with a red curve showing what the distribution would look like if returns were perfectly normally distributed with the same mean and standard deviation.

How to read it:

- The x-axis is daily return in percentage
- The y-axis is density (the area of all bars sums to 1)
- Each bar shows how frequently returns in that range occurred
- The red curve is the theoretical normal distribution fitted to the same mean and variance

What specific things to look for:

Center peak: The blue bars near zero should be significantly taller than the red curve. This means returns cluster around zero more tightly than the normal predicts — the distribution is "pointy" in the center.

Left tail (extreme losses): The blue bars on the far left should extend further than the red curve, or there should be bars where the red curve has essentially gone to zero. These are the fat tails — extreme losses that happen more often than normal predicts.

Right tail: Similarly, large gains occur more often than the normal predicts, but this matters less for risk management.

Good result (expected result):
The empirical histogram visibly exceeds the normal curve at the center and in both tails. The excess kurtosis value in the summary statistics (next output) should be positive and meaningfully above zero.

Warning sign:
If the histogram closely matches the red curve everywhere, it means your portfolio returns are unusually normal — possible, but rare for real financial data. More likely it means the return computation is wrong (perhaps using too long a time window and inadvertently aggregating returns).

Why it matters:
This is the single most important diagnostic chart in the entire project. It is the visual proof that the **parametric VaR assumption is wrong**. When you show this chart in an interview and say "this gap between the red curve and the blue bars in the left tail is exactly why parametric VaR underestimates 99% risk," you are demonstrating real risk intuition. Every subsequent finding — the backtesting failures, the CVaR/VaR ratio analysis — flows from this observation.

OUTPUT 1.7 — Summary Statistics Table

What you are looking at:
A table with one row per asset plus a Portfolio row, and seven columns of statistics.

How to read each column:

Ann. Return (%): The asset's expected yearly gain, annualised from daily returns by multiplying by 252. SPY and QQQ should be the top equity performers. TLT may be negative or near zero over 2015–2024 due to 2022.

Ann. Vol (%): How much the asset's price fluctuates year-to-year. Higher = riskier. Typical ranges:

- SPY: 15–18%
- QQQ: 18–22%
- GLD: 12–15%
- TLT: 12–16%
- VNQ: 18–22%
- Portfolio (equal weight): 10–14%

Sharpe Ratio (rf=0): Return divided by risk. Higher is better. SPY should be around 0.7–1.0 for this period. The portfolio Sharpe should be comparable to or better than most individual assets due to diversification.

Skewness: Negative values (common for equities) mean there are more extreme down days than extreme up days. SPY skewness is typically around -0.5 to -1.0. A strongly negative skew is a warning for VaR models.

Excess Kurtosis: This is the critical number. **Expect values of 2–8 for individual assets, and 2–5 for the portfolio.** Any value above zero confirms fat tails. The higher this number, the more the parametric VaR underestimates tail risk.

Min/Max Daily (%): The single worst and best day in the entire dataset. For SPY, the worst day should be around -10 to -12% (a COVID day in March 2020). This is the empirical bound on single-day losses in your data.

Good result:

- Portfolio Ann. Vol is meaningfully lower than the average Ann. Vol of the 8 individual assets — this proves diversification is reducing risk
- Excess Kurtosis > 2 for equities — confirms fat tails and motivates the backtesting findings
- Negative skewness for equity assets — realistic

Warning sign:

- Portfolio Ann. Vol is equal to or higher than the average individual vol — diversification is not working, possibly due to a correlation calculation error or truly high pairwise correlations
- Excess Kurtosis near 0 for all assets — return distribution is suspiciously normal, check return computation

Why it matters:
The excess kurtosis values in this table are the quantitative justification for everything you do in Phases 3–5. When an interviewer asks "why did you compute CVaR on top of VaR?", the answer starts here: "Because the excess kurtosis of my portfolio returns is X, confirming fat tails that parametric VaR's normality assumption cannot capture."

PLOT 1.8 — Correlation Matrix Heatmap

What you are looking at:
An 8×8 grid where the color and number in each cell shows the correlation between each pair of assets. Colors range from red (-1.0, perfect negative correlation) through yellow (0, no correlation) to green (+1.0, perfect positive correlation).

How to read it:

- Diagonal is always 1.0 (each asset is perfectly correlated with itself)
- Off-diagonal cells: the number is the Pearson correlation coefficient between the two assets
- Symmetric: the cell for (SPY, TLT) is the same as (TLT, SPY)

What specific correlations to expect and explain:

Pair	Expected Value	Why
SPY ↔ QQQ	+0.85 to +0.93	Both US equity; QQQ is a subset of SPY's holdings
SPY ↔ EFA	+0.75 to +0.85	Both equity; different geography but same risk-on/risk-off behavior
SPY ↔ TLT	-0.30 to +0.10	Historical flight-to-safety; may be near zero or slightly positive post-2022
SPY ↔ GLD	-0.10 to +0.15	Gold is an inflation/crisis hedge; near-zero typical correlation
SPY ↔ LQD	+0.40 to +0.65	Both risk assets; LQD has equity-like credit risk
SPY ↔ VNQ	+0.60 to +0.75	REITs are equity-like but also interest rate sensitive
TLT ↔ LQD	+0.60 to +0.80	Both fixed income; both decline when rates rise
GLD ↔ TLT	+0.10 to +0.35	Both "safe haven" assets but different drivers

Good result:
Clear differentiation across the matrix. The upper-left equity block (SPY, QQQ, EFA, XOM, VNQ) should be a warm green cluster. The GLD and TLT rows/columns should show cooler colors against equities.

Warning sign:

- All cells near +0.90 to +1.0 — portfolio is effectively undiversified; you're just holding 8 versions of the same risk. The VaR would then barely benefit from diversification.

- Any correlation of exactly 1.0 for two different assets — data error, likely duplicate column
- TLT showing strongly positive correlation with SPY (+0.5 or higher) — this reflects the post-2022 rate regime and is actually an important real-world finding to mention: "My data covers the 2022 regime where the traditional negative stock-bond correlation broke down"

Why it matters:
The correlation matrix is the input to the covariance matrix, which is the foundation of parametric VaR and Monte Carlo simulation. If your correlations are wrong, all downstream VaR numbers are wrong. More importantly, this matrix determines how much diversification your portfolio actually achieves — low or negative pairwise correlations translate directly into lower portfolio VaR relative to individual asset VaRs.

OUTPUT 1.9 — Portfolio Variance Cross-Check

What you are looking at:
Two numbers printed side by side:

```
Portfolio Annualised Variance : 0.013456
Portfolio Annualised Vol      : 11.597%
```

How to read it:

- Variance is the raw number (σ^2); volatility is its square root expressed as a percentage
- The formula used: $\sigma_p^2 = \mathbf{w}^T \Sigma \mathbf{w}$
- This should match the Portfolio row in the summary statistics table

Good result:
The "Ann. Vol (%)" printed here should match the "Ann. Vol (%)" in the summary statistics table to at least 2 decimal places. If they do not match, there is an error in either the covariance matrix computation or the weight vector.

Warning sign:
Mismatch between this number and the summary statistics table. This would mean your covariance matrix and your direct variance calculation are giving different answers — implying a bug in one of them.

Why it matters:
This is a critical internal validation. The covariance matrix (Σ) is what parametric VaR and Monte Carlo both use. If $\mathbf{w}^T \Sigma \mathbf{w}$ gives a different answer from `portfolio_returns.var() * 252`, the covariance matrix is wrong and all downstream VaR calculations are corrupted.

PLOT 1.10 — Annualised Volatility Bar Chart

What you are looking at:
A bar chart with one bar per asset showing annualised volatility in percentage. A horizontal dashed line marks the portfolio's annualised volatility.

How to read it:

- Bars colored **red** = assets with volatility **above** the portfolio volatility
- Bars colored **blue** = assets with volatility **below** the portfolio volatility
- The dashed line = portfolio volatility (the diversification result)

Good result:
Most bars should be red — meaning most individual assets are more volatile than the portfolio. This is diversification doing its job. The portfolio volatility should be meaningfully below the simple average of all asset volatilities.

Plain English explanation: If you held only SPY, your annual volatility might be 17%. If you held only TLT, it might be 14%. But because SPY and TLT often move in opposite directions, holding both together produces a portfolio volatility of, say, 12% — lower than either asset alone. This chart makes that invisible mathematical benefit visible.

Warning sign:
Portfolio volatility line sits at or above the average bar height — diversification is not working. This would happen if all assets are highly positively correlated. Check the correlation matrix.

Why it matters:
The portfolio volatility is the denominator of the Sharpe ratio and the direct input to parametric VaR. A lower portfolio vol from diversification directly produces a lower (more favorable) parametric VaR estimate. This chart is proof that the portfolio construction is sound before any risk calculation begins.

Phase 2 — Value at Risk

OUTPUT 2.1 — Historical Simulation VaR

What you are looking at:

```
── Historical Simulation VaR ──────────────────────────
  95% VaR : 1.1023%  →  on a $1M portfolio = $11,023
  99% VaR : 1.8912%  →  on a $1M portfolio = $18,912
```

How to read it:

- 95% VaR of 1.10% means: on 95% of trading days, the portfolio will not lose more than 1.10%

- Equivalently: on 5% of days (roughly 1 day per month), losses will exceed 1.10%
- The dollar figure is simply the percentage multiplied by $1,000,000

What the specific number means in plain English:
"Based on the last 252 days of actual market data, the worst case loss we expect to see on any given day — with 99% confidence — is about $18,900. One day in a hundred should exceed this."

Good result / what to expect:

- 95% HS VaR: typically 0.9% – 1.3% for this diversified portfolio
- 99% HS VaR: typically 1.7% – 2.5%
- 99% should be roughly 1.5–2.0x the 95% estimate — the tail should be thicker than a normal distribution would suggest

Warning sign:

- 95% VaR > 99% VaR — mathematically impossible; means a calculation error (percentile sign flip)
- 99% HS VaR very close to 95% HS VaR (within 0.1%) — suggests an extremely thin tail, which is unusual for equity-containing portfolios and may indicate too short a lookback window

Why it matters:
HS VaR is the regulatory baseline — it is the simplest, most auditable method and the first comparison point. Everything else is evaluated relative to this number. If HS says $18,900 and parametric says $14,000, that $4,900 gap is the first evidence that the normality assumption is dangerous at the 99% level.

PLOT 2.2 — HS VaR on Return Distribution

What you are looking at:
The return distribution histogram (same as Plot 1.6) with two vertical dashed lines added: orange for 95% VaR and red for 99% VaR.

How to read it:

- Everything to the LEFT of the orange line represents the worst 5% of days
- Everything to the LEFT of the red line represents the worst 1% of days
- The lines cut the x-axis at the negative of the VaR value (e.g., if 99% VaR is 1.89%, the red line sits at x = -1.89%)

Good result:

- Exactly 5% of the histogram bars are to the left of the orange line (by construction for HS VaR)
- The lines are clearly visible and separated — the 99% line should be noticeably further into the left tail than the 95% line

- The bars beyond the 99% VaR line should still be visible — this is the tail that exceeds VaR, which is what CVaR will measure

Warning sign:

- The two lines are very close together — thin tail, unusual for real financial data
- No bars visible beyond the 99% line — would mean CVaR cannot be computed (no observations in the tail), which happens if the return series is too short

Why it matters:
This chart gives an immediate visual sense of how conservative (or not) the VaR threshold is. Seeing bars extending well beyond the 99% line is the visual argument for CVaR — "VaR tells you where the danger zone starts, but look how much is still out there beyond that line."

OUTPUT 2.3 — Parametric VaR Intermediate Values

What you are looking at:

```
Daily portfolio σ : 0.6542%
z at 95%          : -1.6449
z at 99%          : -2.3263
```

How to read it:

- Daily σ (sigma) is the portfolio's standard deviation on a single day — how much it typically moves
- z-scores are the number of standard deviations below the mean that correspond to each confidence level
- VaR = |z| × σ (since mean is set to zero)

Good result:

- Daily σ of 0.5% – 0.9% is typical for a diversified multi-asset portfolio
- z at 95% should always be -1.6449 (fixed mathematical constant)
- z at 99% should always be -2.3263 (fixed mathematical constant)

Why these z-scores matter in interviews:
Memorise: **-1.645 at 95%, -2.326 at 99%.** These come from `scipy.stats.norm.ppf(0.05)` and `norm.ppf(0.01)`. If someone asks "how do you compute parametric VaR?", the answer is "multiply the portfolio's daily standard deviation by 1.645 (for 95%) or 2.326 (for 99%)."

Warning sign:
z-scores that differ from these constants — they should be fixed numbers from the standard normal table, not estimated from data.

PLOT 2.4 — Parametric VaR: Normal Distribution Overlay

What you are looking at:
Same layout as Plot 2.2 but with additional context: the red bell curve shows the normal distribution assumed by parametric VaR, and the VaR lines now represent the parametric estimates.

Critical thing to look for — the key visual in the entire project:
The parametric VaR line (derived from the normal curve) should sit to the RIGHT of the HS VaR line in the left tail. This means parametric VaR is a smaller (less severe) loss estimate than HS VaR.

Why this happens:
The normal distribution has thin tails. At the 99th percentile, it assigns less probability mass to extreme losses than actually occurred in history. So when you set your VaR threshold using the normal distribution, you are cutting the distribution at a point that is not far enough into the left tail — you are underestimating how bad the worst 1% of days actually are.

Good result (expected result):
Parametric 99% VaR line sits to the right of HS 99% VaR line. This gap is expected and desirable — it demonstrates the cost of the normality assumption and motivates CVaR.

Warning sign:
Parametric VaR line sits further left than HS VaR at every confidence level — would suggest your empirical distribution is actually thinner-tailed than normal, which is unusual for a diversified portfolio. Could indicate a very calm data period was used.

OUTPUT 2.5 — Monte Carlo VaR: Simulation Stats

What you are looking at:

```
Simulated return stats:
  Mean   : -0.0012%
  Std    : 0.6549%
  Min    : -3.1234%
  Max    : +2.9876%
```

How to read it:
These are the statistics of 10,000 simulated daily portfolio returns.

Good result:

- Mean ≈ 0% — by design, since we set $\mu = 0$
- Std should match the Daily σ from parametric VaR (Output 2.3) to within 2–3 decimal places — this is the key convergence check

- Min and Max should be approximately symmetric and roughly ±4–5σ (i.e., very extreme outcomes occur occasionally in 10,000 simulations but not commonly)

Warning sign:
Std of simulated returns deviating significantly (>0.05%) from the analytical portfolio daily σ — means Cholesky decomposition or the simulation mechanics contain an error.

Why it matters:
This is the internal validation for Monte Carlo. If the simulated distribution does not match the analytical one (same mean and variance), the simulation is wrong and the MC VaR will disagree with parametric VaR for the wrong reasons.

OUTPUT 2.6 — Cholesky Reconstruction Error

What you are looking at:

```
Max reconstruction error (L @ L.T - Σ): 2.31e-17  (should be ~0)
```

How to read it:
After computing L = Cholesky decomposition of the covariance matrix, we verify that LL^T reconstructs the original matrix. The max absolute difference across all cells should be essentially zero (machine precision).

Good result:
Error in the range of 10^{-16} to 10^{-18} — this is floating point machine epsilon, mathematically equivalent to zero.

Warning sign:
Error larger than 10^{-10} — suggests numerical instability in the covariance matrix, possibly because one or more asset pairs have correlations very close to +1.0 (near-singular matrix). If Cholesky fails entirely with a "matrix is not positive definite" error, it means two assets are nearly perfectly correlated and one should be removed.

Why it matters:
The Cholesky matrix is used to inject the correct correlation structure into 10,000 Monte Carlo simulations. If it reconstructs the covariance matrix incorrectly, all 10,000 simulations have wrong correlations, making the MC VaR meaningless.

OUTPUT 2.7 — VaR Comparison Table

What you are looking at:

	Historical Sim (%)	Parametric (%)	Monte Carlo (%)
95% VaR	1.1023	1.0124	1.0231
99% VaR	1.8912	1.4287	1.4356

How to read it:
Each row is a confidence level. Each column is a VaR method. The numbers are percentage losses.

The key pattern to explain:

At 95%: All three numbers should be relatively close — within 0.2% of each other. This is expected because the 95th percentile of the empirical distribution is reasonably well-approximated by a normal distribution.

At 99%: A visible gap should open up. HS VaR should be meaningfully larger than parametric and MC VaR. Parametric and MC should be very close to each other (both assume normality).

Good result (expected pattern):

- 95% level: within ±0.2% across all three methods
- 99% level: HS VaR is 0.3% - 0.6% higher than parametric VaR
- MC VaR within 0.05% of parametric VaR (Monte Carlo sampling noise)

Warning sign:

- HS VaR < Parametric VaR at 99% — would mean your historical data has thinner tails than normal, very unusual
- MC VaR differs from parametric by more than 0.1% — simulation bug
- All three methods produce identical numbers — suggests only one method is actually running and the others are copying its result

Why it matters:
This table is the core quantitative finding of Phase 2. The divergence between HS and parametric at 99% quantifies the model risk embedded in the normality assumption. A $4,000–$6,000 difference per $1M portfolio might seem small, but scaled to a $100M fund it becomes $400K–$600K in underestimated risk.

PLOT 2.8 — VaR Method Comparison Bar Chart

What you are looking at:
Two bar charts side by side — one for 95% confidence, one for 99% confidence. Each chart has three bars representing the three VaR methods.

How to read it:

- Taller bar = higher estimated VaR = more conservative (more risk captured)

- The bar heights should diverge more in the 99% chart than in the 95% chart

Good result:

- 95% chart: three bars roughly the same height
- 99% chart: HS bar visibly taller than parametric and MC bars; parametric and MC bars nearly identical

What to say about this chart in an interview:
"The 95% chart shows that all three methods agree reasonably well — the distribution is approximately normal near the middle. But at 99%, historical simulation estimates a significantly larger loss than the parametric model. That gap is the cost of assuming normality. The parametric model, by fitting a bell curve, doesn't know that the actual worst 1% of days are worse than a bell curve would predict."

Phase 3 — CVaR / Expected Shortfall

PLOT 3.1 — CVaR Intuition Chart

What you are looking at:
The return distribution histogram with two vertical lines: orange dashed (VaR threshold) and red solid (CVaR). The area beyond the orange line is shaded red.

How to read it:

- Orange line: the VaR threshold — the point where the worst 5% of days begin
- Red shaded region: all the days that were worse than VaR
- Red solid line: the average of all those red-shaded days — this is CVaR

Key relationship to explain:
CVaR is always to the LEFT of VaR (a larger loss). This must always be true — CVaR is the average of losses that are all worse than VaR, so the average must itself be worse than VaR.

Good result:
Clear visual separation between the VaR line and the CVaR line. A larger separation indicates a heavier tail — when things go wrong, they go very wrong.

Why it matters:
This is the most intuitive way to explain CVaR to a non-technical audience. In an interview with a risk committee member (non-quantitative): "VaR says we lose at most $18,900 on 99% of days. CVaR says that on the 1% of days when we do exceed that, we lose $32,000 on average. VaR tells you where the danger zone starts. CVaR tells you how deep it goes."

OUTPUT 3.2 — CVaR with Tail Observation Count

What you are looking at:

```
95% | VaR = 1.1023%  | CVaR = 1.5821%  | Tail obs = 113  | CVaR/VaR ratio
= 1.436x
99% | VaR = 1.8912%  | CVaR = 3.1845%  | Tail obs = 22   | CVaR/VaR ratio
= 1.684x
```

How to read each element:

Tail obs: The number of historical days that exceeded the VaR threshold. At 95% over ~2,265 days, expect ~113 observations. At 99%, expect ~23. CVaR is computed as the average of these observations.

CVaR/VaR ratio: How much worse is the average tail loss than the VaR threshold. A ratio of 1.68x means that when losses exceed the 99% VaR, they are on average 68% larger than the VaR itself.

Good result (what to expect):

- 95% CVaR/VaR ratio: 1.25 to 1.50 (theoretical normal ratio is ~1.26)
- 99% CVaR/VaR ratio: 1.30 to 1.80 (theoretical normal ratio is ~1.16)
- The empirical ratio should exceed the theoretical normal ratio — if it does not, your tails are thinner than normal, which is unusual

Warning sign:

- Tail obs = 0 — cannot compute CVaR. This would happen with too short a data series or a bug in the exception identification. CVaR would return NaN.
- CVaR/VaR ratio < 1.0 — mathematically impossible. CVaR must always exceed VaR.
- CVaR/VaR ratio at 99% lower than at 95% — also impossible (the deeper tail should always be relatively heavier). Indicates a computation error.

Why the ratio matters:
The ratio directly quantifies the "hidden risk beyond VaR." A fund that reports only VaR to its investors is hiding the CVaR/VaR ratio gap. Regulators moved to Expected Shortfall precisely because this ratio can be very large — in 2008 it was enormous for mortgage-backed security portfolios.

OUTPUT 3.3 — Parametric CVaR Intermediate Values

What you are looking at:

```
99% | VaR = 1.4287%  | CVaR = 1.6589%  | φ(z) = 0.0267  | CVaR/VaR ratio =
1.161x
```

What φ(z) = 0.0267 means:
φ(z) is the standard normal probability density function evaluated at the 99th percentile z-score (-2.326). This number comes from the bell curve's height at the point corresponding to the VaR threshold.

Why the parametric CVaR/VaR ratio is lower than HS:
The parametric method assumes normality. Under a normal distribution, the CVaR/VaR ratio at 99% is theoretically ~1.16. The empirical (HS) ratio will typically be 1.4–1.7. This gap between ~1.16 (normal assumption) and ~1.6 (empirical) is the model risk from assuming normality.

Good result:
Parametric CVaR/VaR ratio close to the theoretical values of 1.26 (95%) and 1.16 (99%). HS CVaR/VaR ratio noticeably higher than these benchmarks.

What to say in an interview:
"The theoretical CVaR/VaR ratio under a normal distribution is 1.16 at 99%. My historical simulation gives a ratio of 1.68. The difference — 0.52 — is entirely explained by fat tails that the normal distribution does not capture. That's the reason I built three methods and computed both VaR and CVaR."

PLOT 3.4 — VaR vs CVaR Bar Chart (Side by Side)

What you are looking at:
Two panels, one for each confidence level. In each panel, three pairs of bars: blue (VaR) and red (CVaR) for each method.

How to read it:
Within each pair, the red bar must always be taller than the blue bar. The difference in height is the additional expected loss once VaR is breached.

Key patterns to identify:

At 95%: The three VaR bars should be similar heights; CVaR bars add roughly 30–50% on top.

At 99%: The HS VaR bar is already taller than parametric/MC VaR bars. The HS CVaR bar should be dramatically taller than the parametric CVaR bar — this is where the fat tail problem is most visible.

Good result:
The HS CVaR at 99% should be the tallest bar in the chart. The gap between HS CVaR and parametric CVaR should be much larger than the corresponding gap at 95%.

Why it matters:
This chart makes the fat tail problem quantitatively concrete. A risk manager relying only on parametric CVaR at 99% might think the danger zone averages $16,000. The historical simulation says $32,000.

That 2x difference in capital allocation is the practical consequence of an incorrect distributional assumption.

Phase 4 — Backtesting

OUTPUT 4.1 — Exception Counts

What you are looking at:

```
HS         | Exceptions:  35 / 2263 | Actual rate: 1.547% | Expected: 22.6 (1.000%)
Parametric | Exceptions:  50 / 2263 | Actual rate: 2.210% | Expected: 22.6 (1.000%)
MC         | Exceptions:  49 / 2263 | Actual rate: 2.166% | Expected: 22.6 (1.000%)
```

How to read it:

- 2,263 = total rolling window observations (all days after the 252-day warmup)
- Expected at 99%: 1% × 2,263 = 22.6 exceptions
- HS: 35 actual vs 22.6 expected → 55% more exceptions than expected
- Parametric: 50 actual vs 22.6 expected → 121% more exceptions than expected

What each number means in plain English:
At 99% confidence, the model promises that losses will exceed VaR on only 1% of days — roughly 23 days out of 2,263. If the model is seeing 50 such days, it is massively underestimating risk. Parametric VaR is saying "this loss level will only be exceeded 23 times" and reality is saying "wrong — 50 times."

Good result (model passes):
Exception count within a reasonable range of 22.6 — Kupiec test will tell you statistically what "reasonable" means, but roughly 15–35 exceptions at 99% over this sample would be consistent with correct calibration.

Warning sign:

- Exceptions far above expected (like 50) → model too optimistic, underestimates risk
- Exceptions far below expected (like 5) → model too pessimistic, over-estimates risk, leading to excess capital held
- Parametric and MC having the same (or very similar) exception counts — expected and correct, since both assume normality

- HS having fewer exceptions than parametric — very unusual, would suggest the historical period was unusually calm relative to recent conditions

---

PLOT 4.2 — Rolling VaR vs Actual Returns

What you are looking at:
Three stacked panels, one per VaR method. In each panel: grey line = actual daily portfolio return; colored line = the negative of the rolling VaR threshold (the predicted "floor"); red dots = exception days.

How to read it:

- The colored line is not the VaR number itself — it is **−VaR**, so it plots below zero as a loss threshold
- On any day where the grey line dips below the colored line, that's an exception (marked by a red dot)
- The VaR line should move over time as it is recalibrated every day from the rolling 252-day window

What specific patterns to look for:

Pre-2020 calm period: The VaR line should be relatively flat and close to zero — a calm market calibrates a low VaR. Few or no red dots.

March 2020: A cluster of red dots in a very short window. The actual returns (grey line) should spike sharply downward, passing through the VaR line repeatedly. This is the exception clustering the Christoffersen test detects.

After March 2020: The VaR line should shift downward (more negative) as the volatile March data enters the rolling window. This is the model "catching up" — it only updates after it has already missed the exceptions.

2022: Similar pattern — cluster of exceptions, then VaR line adapts.

Good result:
The VaR line tracks roughly 1–2% below zero in calm periods and deepens to 2–3% during volatile periods. Red dots should be scattered (at most 1–2 per year in a good model) with some inevitable clustering during genuine crises.

Warning sign:

- VaR line flat throughout — means the rolling window computation is not actually rolling (a static VaR mistakenly applied). This is a code error.
- Every single exception (red dot) appears in one short window and nowhere else — extreme clustering, the Christoffersen test will strongly reject independence

Why it matters:
This plot directly shows the fundamental failure mode of static rolling-window VaR: it is **backward-**

looking and slow to respond. The model only increases its VaR estimate after it has already experienced the exceptions it failed to predict. A GARCH model would predict the volatility spike before it happens.

PLOT 4.3 — Exception Clustering Timeline

What you are looking at:
Three horizontal timelines (one per method) showing vertical tick marks at every exception date.

How to read it:

- Each vertical line = one day where actual loss exceeded VaR
- Dense clusters of vertical lines = exceptions are happening back-to-back
- Isolated single lines = exceptions are randomly scattered

What to look for:

- **Early 2020:** Dense cluster of lines — the COVID crash
- **Late 2022:** Another cluster — the rate shock
- **2016–2019 and 2021:** Sparse, isolated lines — the calm periods

Good result (for the model):
Lines evenly scattered throughout the timeline with no visible clusters. This would mean exceptions are independent — the model is capturing volatility dynamics adequately.

Expected (real) result:
Clear clusters in 2020 and 2022. This is what causes the Christoffersen test to fail.

What to say about this chart:
"You can see that exceptions don't happen randomly — they come in waves. In March 2020, the model was breached many days in a row. A good model would not produce this pattern; it would know that one bad day makes the next bad day more likely, and it would raise the VaR estimate accordingly. Rolling historical simulation is blind to this. You'd need GARCH to fix it."

OUTPUT 4.4 — Kupiec Test Results

What you are looking at:

Method	N	T	p_hat	p_expected	LR_uc	p-value	Result
HS	35	2263	1.547%	1.000%	6.234	0.0126	FAIL ✗
Parametric	50	2263	2.210%	1.000%	26.847	0.0000	FAIL ✗
MC	49	2263	2.166%	1.000%	25.213	0.0000	FAIL ✗

How to read each column:

N: Number of exceptions observed.

T: Total rolling observations.

p_hat: Actual exception rate = N/T.

p_expected: What the model promises (1% for 99% VaR).

LR_uc: The likelihood ratio test statistic. Reject the null (model fails) if this exceeds **3.841** (chi-squared critical value with 1 degree of freedom at 5% significance).

p-value: Probability of observing this many exceptions if the model were correctly specified. A p-value of 0.0126 means there is only a 1.26% chance of seeing 35+ exceptions if the true rate were 1%. Since 1.26% < 5%, we reject — the model fails.

Result:

- PASS ✓ = p-value ≥ 0.05: exception rate is statistically consistent with model
- FAIL ✗ = p-value < 0.05: exception rate is inconsistent with model

Good result (for 95% VaR):
All three methods should PASS — exception rates of 4.6–5.4% are statistically consistent with the promised 5%.

Expected result (for 99% VaR):
All three methods FAIL — exception rates of 1.5–2.2% are inconsistent with the promised 1%, with parametric being the worst offender.

What to say about the parametric failure at 99%:
"Parametric VaR fails the Kupiec test with a p-value near zero — it's not even close. The model promised that losses would exceed VaR on 1% of days, but it happened on 2.2% of days. That's more than double the promised rate. The reason is the normality assumption: the normal distribution assigns too little probability to extreme losses, so the VaR threshold is set too conservatively and gets breached far too often."

OUTPUT 4.5 — Christoffersen Test Results

What you are looking at:

```
HS
  Transition counts   : n00=2192  n01=  36  n10=  35  n11=   0
  π01 (exc | no exc)  : 0.0161
  π11 (exc | exc)     : 0.0000  ← should ≈ π01 if independent
  LR_ind              : 8.432   p=0.0037  FAIL ✗
  LR_cc (full test)   : 14.666  p=0.0007  FAIL ✗
```

How to read the transition matrix:

- **n00 = 2192:** Days where yesterday had no exception AND today has no exception. The normal case — 97% of all days.
- **n01 = 36:** Days where yesterday had no exception BUT today does. A "fresh" exception starting.
- **n10 = 35:** Days where yesterday had an exception BUT today does not. An exception ending.
- **n11 = 0:** Days where yesterday had an exception AND today also does. Consecutive exceptions.

π01 = 0.0161: Given that yesterday was a normal day, there is a 1.61% chance of an exception today.

π11 = 0.0000: Given that yesterday was an exception, there is a 0% chance of an exception today.

Is this clustering or not?
If exceptions were perfectly independent, π01 should equal π11. Here π11 = 0 < π01 = 1.61%, suggesting exceptions do not follow exceptions. However — this is a sample-size artifact. With only 35 exceptions out of 2,263 days, having n11 = 0 is plausible even with clustering, because exceptions are rare.

The LR_ind statistic tests this formally. A p-value of 0.0037 means the independence null is rejected at 5% significance — exceptions are not independent.

What to say about this result:
"Even though n11 = 0 (no consecutive exception days in the HS model), the Christoffersen test rejects independence because the timing of exceptions is not consistent with a random 1.6% daily probability. The exceptions come in bursts — multiple exceptions within a short window — rather than being uniformly distributed across the 10-year sample."

Good result:
LR_ind p-value > 0.05 → exceptions are approximately independent → model captures volatility dynamics

Expected (real) result:
LR_ind p-value < 0.05 for all methods → exceptions cluster → model is blind to volatility regimes

---

OUTPUT 4.6 — Combined Backtesting Summary Table

What you are looking at:
A table combining Kupiec and Christoffersen results for all three methods at a single confidence level, with PASS/FAIL for both tests.

The four possible outcomes for a VaR model:

Kupiec	Christoffersen	What it means
PASS	PASS	Model is well-specified — correct frequency and independent exceptions

Kupiec	Christoffersen	What it means
PASS	FAIL	Frequency is right but exceptions cluster — model misses volatility regimes
FAIL	PASS	Wrong frequency but random — model is mis-calibrated but at least consistent
FAIL	FAIL	Both wrong — model is fundamentally mis-specified

Your actual results:

- 95% VaR: PASS / FAIL (frequency ok, clustering present)
- 99% VaR: FAIL / FAIL (model fundamentally wrong in the deep tail)

What to say:
"The 95% results tell me the average calibration is acceptable — the model gets the right number of exceptions over a long period. But the Christoffersen failure tells me it gets them in the wrong order — clustered during crises. The 99% results tell me the model is fundamentally wrong in the deep tail; it's not just a timing problem, it's a distributional problem. This is the core motivation for stress testing."

Phase 5 — Stress Testing

PLOT 5.1 — Individual Crisis Drawdown Paths

What you are looking at (COVID example):
A time series chart of the portfolio's cumulative log return from the start of the COVID crash (Day 0 = Feb 20, 2020) to the market bottom (Day 23 = Mar 23, 2020).

How to read it:

- x-axis: trading days from crisis start (all three scenarios start at Day 0 for comparability)
- y-axis: cumulative portfolio loss in percentage (negative values = losses)
- The curve shows how the portfolio's value declined day by day

COVID (orange):
Near-vertical drop. The cumulative loss line should plunge to around -25% to -32% within 23 trading days. This represents one of the fastest drawdowns in market history.

GFC (red):
Straight line declining gradually over 126 days to -28% to -35%. (Note: GFC uses simulated equal daily shocks, hence the straight line rather than the jagged actual path.)

Rate Shock (purple):
Long, jagged decline over ~200 days. The cumulative loss will reach -18% to -25%. Unlike COVID, there are partial recovery periods along the way, but the overall direction is down.

Reference lines:

- **Blue dashed:** 99% VaR (~-1.9%) — this is the model's "maximum expected loss" on a normal day
- **Red solid:** 99% CVaR (~-3.2%) — the average tail loss

Key observation:
All three crisis paths breach both reference lines within the first 1–3 trading days and never recover above them during the crisis. The model's "worst case" is surpassed almost immediately when a real crisis begins.

Good result (expected):
All three paths sit well below both reference lines throughout. The larger the gap between the crisis paths and the reference lines, the more clearly the project demonstrates the limitations of statistical VaR.

Warning sign:
If crisis paths barely breach the VaR line, it means either the VaR is set too conservatively (too high) or the stress scenarios are too mild.

OUTPUT 5.2 — Per-Scenario Stress Metrics

What you are looking at:

```
COVID Stress Metrics:
  Total loss (window)  : -29.45%  → $-294,500
  Peak drawdown        : -29.45%  → $-294,500
  Worst single day     : -7.23%   → $-72,300
  Avg daily loss       : -1.45%   → $-14,500

  99% HS VaR (1-day)   : 1.89%   → $18,900
  Worst COVID day vs VaR: 3.82x the 99% VaR
```

How to read the "vs VaR" multiple:
The worst COVID day produced a loss 3.82 times larger than the 99% VaR estimate. This means the model would have to be breached nearly 4 times simultaneously to equal one worst-day COVID loss.

Plain English:
"On the worst day of the COVID crash, the portfolio lost $72,300. Our statistical model said the worst 1% of days would cost $18,900. The actual worst COVID day was nearly 4 times larger than what the model considered a 1-in-100 event."

Good result (in terms of storytelling):

- "vs VaR" multiple of 2x–5x for COVID and GFC — confirms VaR drastically underestimates crisis losses
- Avg daily loss during COVID significantly exceeding CVaR — confirms that during sustained crises, even CVaR underestimates

Why the Rate Shock analysis is especially important:
The per-asset breakdown during the Rate Shock should show TLT and LQD (bond ETFs) posting large negative returns simultaneously with SPY (equity). This directly contradicts the negative stock-bond correlation assumed in the covariance matrix. Say: "In 2022, my diversification assumption broke down. The covariance matrix calibrated on 2015–2021 data embedded a negative stock-bond correlation, but in 2022 both equities and bonds fell simultaneously. This is correlation risk — the risk that diversification fails exactly when you need it most."

OUTPUT 5.3 — Hypothetical Scenario Loss

What you are looking at:

```
Portfolio loss          : 13.45%
Dollar loss ($1M)       : $134,500
vs 99% VaR              : 7.11x
vs 99% CVaR             : 4.22x
```

How to read it:
A single-day hypothetical liquidity crisis would produce a 13.45% portfolio loss — 7x the 99% VaR and 4x the 99% CVaR.

Why this scenario is included:
Historical scenarios can only capture past events. The hypothetical scenario represents a stress test designed by the risk analyst — forward-looking and scenario-specific. This is how bank stress tests (DFAST, CCAR) and regulatory exercises (EBA stress tests) are designed. Real risk management is not just "replay 2008" — it is "design the scenario that could break the portfolio next time."

PLOT 5.4 — Stress Test Bar Chart (Key Chart)

What you are looking at:
Four bars representing worst-day losses for each scenario, with three horizontal reference lines: 95% VaR (green dashed), 99% VaR (blue dashed), and 99% CVaR (red solid).

How to read it:

- Each bar shows how much the portfolio would lose on the worst day of that scenario, in dollar terms on a $1M portfolio
- The reference lines show where the statistical model's risk thresholds sit
- Bar labels show the dollar amount

The pattern you should see:
Every single bar clears every single reference line comfortably. The lowest bar (Rate Shock worst day) should still be 2–3x the 99% CVaR line.

This chart is the visual proof of the entire project's thesis: Statistical models (VaR, CVaR) are necessary but not sufficient. They work for normal market conditions. They fail systematically during crises. Stress testing provides the crisis-level risk picture that statistics cannot.

Good result:
All bars substantially above all reference lines. Larger gaps make the point more forcefully.

Warning sign:
If any bar falls below the 99% CVaR line — would mean the stress scenario is actually mild by historical standards and not a genuine stress test. Design a more severe scenario.

What to say in an interview:
"Every stress scenario I tested produced losses that would blow through both the 99% VaR and 99% CVaR in a single day. This is not a failure of the portfolio — it's a fundamental feature of market crises that statistical models calibrated on normal-period data cannot capture. VaR should be the daily risk limit. Stress testing is what tells you how much additional capital buffer you need for tail events."

Phase 6 — Risk Dashboard

DASHBOARD — Reading the Complete Picture

The dashboard consolidates all findings. Here is how to walk someone through it panel by panel in under 5 minutes:

Start with Panel 1 (Cumulative Return + Drawdown):
"This is what the portfolio actually did. The blue line is our cumulative return — overall the portfolio grew over 2015–2024. The red shading is the drawdown — how far we were below our all-time high at any point. You can see two major drawdown periods: a sharp but quick one in 2020, and a prolonged one in 2022."

Move to Panel 3 (Rolling VaR vs Returns):
"This is where the risk model gets tested in real time. The colored lines are the model's daily predictions

of the worst expected loss. Every red dot is a day the model was wrong — actual losses exceeded the prediction. Notice how red dots cluster in 2020 and 2022 — the model couldn't predict those crises."

Then Panel 4 (Exception Timeline):
"This makes the clustering pattern undeniable. These vertical lines are the exception days. They don't appear randomly — they come in dense bursts. That's what the Christoffersen test catches statistically."

Then Panel 5 (Stress Test Bar Chart):
"Even setting aside the statistical failures, look at what happens during real crises. Every crisis scenario produces losses that blow through both VaR and CVaR within a single day. VaR is a useful daily risk limit, but it is not a crisis management tool."

Finish with Panel 6 (Backtesting Table):
"The formal test results confirm everything we saw visually. Green means the model passed, red means it failed. The 95% results are mixed — frequency is right, but timing is wrong. The 99% results are entirely red — the model fails in every dimension in the deep tail."

Master Summary Table

Phase	Key Output	Expected / Good Value	Warning Sign
1	Rows after dropna	~2,500 rows, <10 dropped	>50 rows dropped → data quality issue
1	Days with |return| > 20%	0 for all assets	Any non-zero → data error, investigate
1	Excess Kurtosis (portfolio)	2.0 – 5.0	Near 0 → suspiciously normal; >10 → data error
1	Portfolio Ann. Vol	10% – 14%	> avg individual vol → diversification failing
1	Covariance cross-check	Matches Ann. Vol to 2 decimal places	Mismatch → covariance matrix error
1	Cholesky reconstruction error	< 1e-10	> 1e-10 → near-singular covariance matrix
1	SPY ↔ TLT correlation	-0.30 to +0.10	> +0.30 → stock-bond correlation regime shift (flag it)
2	HS VaR 95%	0.9% – 1.3%	Outside range → check lookback window
2	HS VaR 99%	1.7% – 2.5%	< 95% VaR → sign error in calculation
2	HS vs Parametric VaR gap at 99%	0.3% – 0.6%	Gap < 0.1% → no fat tail evidence (unusual)

Phase	Key Output	Expected / Good Value	Warning Sign
2	MC vs Parametric VaR difference	< 0.05%	> 0.1% → MC simulation bug
2	Simulated return Std	Matches daily σ to 2 decimal places	Large mismatch → Cholesky error
3	HS CVaR/VaR at 95%	1.25 – 1.50 (theoretical normal: 1.26)	< 1.0 → impossible, computation error
3	HS CVaR/VaR at 99%	1.30 – 1.80 (theoretical normal: 1.16)	Equal to parametric ratio → no fat tail
3	Tail observation count at 99%	~22–25 observations	0 → cannot compute CVaR, data too short
4	Exception count at 95% (HS)	~113 out of ~2,263 (5%)	> 150 or < 80 → model mis-calibrated
4	Exception count at 99% (HS)	~22–35 out of ~2,263	> 45 → strong model failure
4	Parametric exceptions at 99%	~40–55 (expected failure)	< 30 → parametric performing too well, check data
4	Kupiec p-value at 95%	> 0.05 (PASS)	< 0.05 → exception frequency wrong
4	Kupiec p-value at 99%	< 0.05 (FAIL — expected)	> 0.05 → surprising pass, verify computation
4	Christoffersen p-value	< 0.05 (FAIL — expected at both levels)	> 0.05 → exceptions are independent (unusual for rolling VaR)
4	π11 vs π01	π11 > π01 indicates clustering	π11 = π01 → no clustering → model captures volatility dynamics
5	COVID worst day vs 99% VaR	3x – 5x	< 1.5x → stress scenario too mild
5	All scenarios vs CVaR	All bars above CVaR line	Any bar below CVaR → scenario not extreme enough
5	Rate Shock: TLT and LQD returns	Both negative simultaneously	Both positive → 2022 correlation regime not captured
5	GFC portfolio total loss	-25% to -40%	< -15% → shock vector too mild; > -60% → check weights
6	Dashboard renders without error	All 6 panels visible, tables formatted	Missing panels → pickle key error from earlier phase
6	Backtesting table colors	95%: mixed; 99%: all red	All green → VaR model suspiciously good; check computation

End of Interpretation Guide

Related document: Technical Documentation (separate PDF)
Notebook files: `phase1.ipynb` through `phase6.ipynb`